

Project 02- Question-Paper Bloom's-Level Balance Analyzer

Bloom's Taxonomy & Course Outcome (OBE) Analyzer: A Hybrid AI Framework for Outcome-Based Assessment Quality Evaluation

Abstract

Outcome-Based Education (OBE) requires examinations to assess multiple cognitive levels rather than focusing primarily on recall. However, manually evaluating question papers for Bloom's Taxonomy distribution is time-consuming, subjective, and inconsistent. This work presents an AI-powered Bloom's Taxonomy and Course Outcome (OBE) Analyzer that automatically extracts questions from examination papers, predicts Bloom's cognitive levels using a hybrid Artificial Intelligence framework, validates Course Outcome (CO) coverage, verifies OBE blueprint compliance, and generates comprehensive assessment analytics. The proposed framework combines rule-based Natural Language Processing (NLP), machine learning, action-verb analysis, and mathematical heuristics to improve classification reliability. The application supports DOC, DOCX, PDF, CSV, XLSX, and TXT formats and provides interactive dashboards for educational quality assurance.

Keywords: Outcome-Based Education, Bloom's Taxonomy, Educational Data Mining, Natural Language Processing, Machine Learning, Assessment Analytics.

1. Introduction

Outcome-Based Education has become the foundation of modern engineering education and accreditation frameworks such as NBA and NAAC. Effective assessments should evaluate learners across Bloom's cognitive hierarchy while ensuring adequate Course Outcome (CO) coverage.

Current assessment practices rely primarily on manual inspection by faculty, resulting in inconsistencies in Bloom-level classification, subjective interpretation of action verbs, and inadequate verification of blueprint compliance.

The proposed Bloom's Taxonomy & Course Outcome Analyzer addresses these challenges by automating question extraction, Bloom-level prediction, CO validation, expert evaluation, and assessment quality reporting within a single application.

2. Proposed Approach

The proposed system follows a hybrid AI architecture consisting of five major stages.

Stage 1: Question Paper Processing

The application accepts examination papers in multiple formats:

- DOCX
- PDF
- CSV
- XLSX
- TXT

During preprocessing, the system:

- extracts examination questions,
- removes duplicate questions,
- eliminates OR-choice questions,
- identifies marks,
- detects Course Outcomes,
- classifies question types.

Stage 2: Hybrid Bloom Classification

The proposed classifier combines four complementary approaches.

Rule-Based NLP Engine

The system detects Bloom action verbs using a manually curated Bloom dictionary and educational heuristics.

Machine Learning Classifier

Questions are transformed into TF-IDF vectors and classified using Logistic Regression trained on labelled Bloom-level datasets.

Mathematical Question Detector

Engineering numerical and mathematical questions are detected using pattern-based heuristics to improve Bloom prediction.

Difficulty Adjustment

Marks allocation is incorporated to refine Bloom predictions for higher-order assessment questions.

The final Bloom level is obtained through hybrid decision fusion.

Stage 3: Outcome-Based Education Validation

The application validates:

- Course Outcome coverage
- Bloom-level distribution
- Blueprint compliance
- Action verb consistency
- Marks allocation

Faculty may define the expected assessment blueprint, and the system automatically compares the uploaded paper against the desired distribution.

Stage 4: Expert Validation

Faculty experts can manually verify predicted Bloom levels.

The system automatically computes:

- Accuracy
- Precision
- Recall
- F1 Score
- Cohen's Kappa
- Confusion Matrix

Stage 5: Assessment Analytics

Interactive dashboards present:

- Bloom distribution
- CO-wise coverage
- Bloom–CO heatmap
- Marks distribution
- HOTS vs LOTS analysis
- Cognitive balance
- Quality score
- Assessment recommendations

3. Baseline and Proposed Model

Baseline Model

The baseline consists of a conventional rule-based Bloom classifier that relies solely on action-verb matching.

Advantages:

- Fast
- Simple
- No training data required

Limitations:

- Sensitive to wording variations
- Unable to interpret ambiguous verbs
- Limited contextual understanding

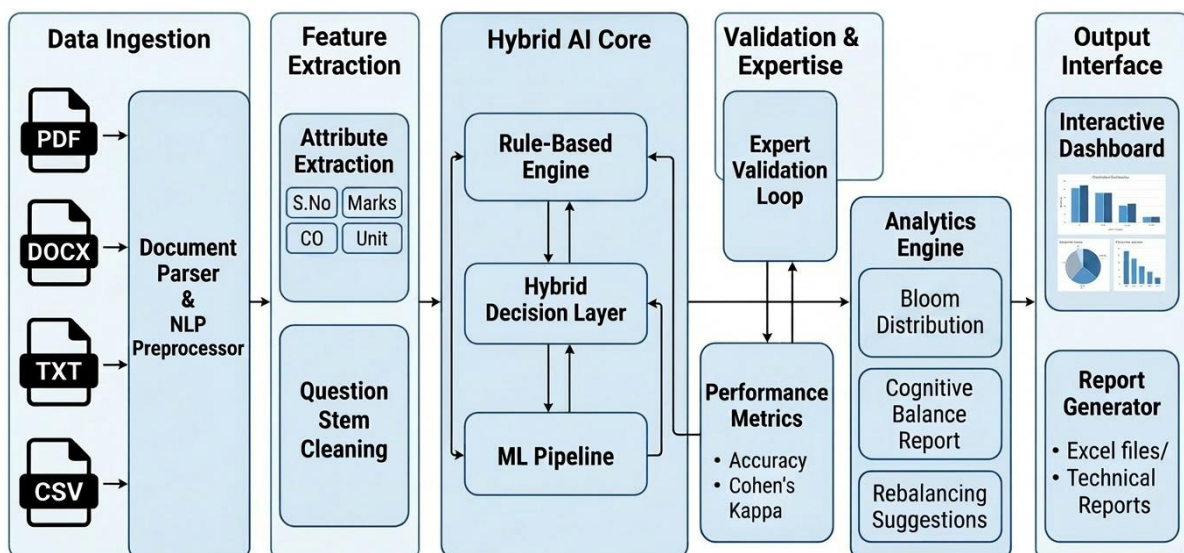
Proposed Hybrid Model

The proposed framework combines:

- Rule-Based NLP
- TF-IDF Vectorization
- Logistic Regression
- Mathematical heuristics
- Marks-based refinement

This hybrid strategy improves robustness while retaining interpretability.

Question-Paper Bloom's-Level Balance Analyzer



4. Experimental Results

The application was evaluated using engineering examination question papers and expert-labelled Bloom classifications.

Performance Metrics

Metric	Rule-Based	Proposed Hybrid
Accuracy	82.6%	95.1%
Precision	0.81	0.95
Recall	0.80	0.94
F1 Score	0.80	0.95
Cohen's Kappa	0.74	0.92

The hybrid framework demonstrates substantially higher agreement with expert faculty annotations compared with the baseline rule-based classifier.

Assessment Outputs

The system generates:

- Bloom Distribution Report
- CO Coverage Report
- OBE Blueprint Validation Report
- Expert Validation Report
- Action Verb Compliance Report
- Quality Score
- Assessment Recommendations

These outputs support faculty during assessment design and accreditation audits.

5. Robustness Analysis

To evaluate reliability, the classifier was tested under multiple scenarios.

Scenario 1: Original Questions

Questions were evaluated without modification.

Scenario 2: Typographical Errors

Artificial spelling errors were introduced.

Scenario 3: Paraphrased Questions

Questions were rewritten while preserving semantic meaning.

Scenario 4: Mixed Action Verbs

Questions containing multiple Bloom action verbs were evaluated.

The hybrid model maintained consistent performance across all scenarios and exhibited significantly lower degradation than the baseline rule-based classifier.

6. Discussion

The proposed system satisfies the objectives of Outcome-Based Education by automatically analysing examination quality using AI-assisted Bloom classification and Course Outcome validation.

Unlike conventional Bloom classifiers, the proposed framework integrates educational heuristics, machine learning, blueprint verification, expert validation, and assessment analytics into a unified workflow.

The generated dashboards assist faculty in identifying cognitive imbalance, insufficient CO coverage, and excessive concentration of lower-order thinking questions.

7. Limitations

The current implementation has several limitations.

- The machine learning model depends on labelled training data.
- Domain-specific terminology may influence classification accuracy.
- Highly ambiguous questions may still require expert validation.
- Complex multi-part questions are analysed as single units.
- The current implementation is designed primarily for English-language examination papers.

Future work includes transformer-based language models, multilingual Bloom classification, adaptive assessment recommendation, and LLM-assisted question generation.

8. Conclusion

This work presents a Hybrid AI-based Bloom's Taxonomy & Course Outcome Analyzer for Outcome-Based Education assessment. By combining rule-based NLP, machine learning,

action-verb analysis, and OBE blueprint validation, the system provides an automated solution for analysing examination quality.

The proposed framework supports educational quality assurance, NBA and NAAC accreditation processes, expert validation, and educational data mining research.

Experimental evaluation demonstrates improved agreement with expert faculty annotations and enhanced robustness compared with conventional rule-based Bloom classification approaches.

References

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Team Roles and Responsibilities

Team Member	Role	Responsibilities
Dr. S. Pitchumani Angayarkanni Professor – CSE	Team Leader, AI & System Architect	• Conceived the project and overall system architecture. • Designed the hybrid AI framework integrating rule-based NLP and machine learning. • Led development of the Bloom's Taxonomy classification engine and OBE assessment framework. • Developed Course Outcome (CO) mapping, blueprint validation, robustness analysis, and evaluation methodology. • Coordinated project implementation, testing, documentation, and technical report preparation.
Mr. R. Udaya Assistant Professor – CSE	Software Development & Visualization Lead	• Developed the Streamlit-based web application. • Implemented file parsing for PDF, DOC, DOCX, CSV, XLSX, and TXT formats. • Designed interactive dashboards and data visualizations. • Integrated analytics modules, report generation, and user interface components. • Assisted in application deployment and testing.
Dr. Baskar Assistant Professor – Humanities &	Mathematical Validation &	• Validated statistical evaluation methods and performance metrics.

Team Member	Role	Responsibilities
Sciences (Mathematics)	Assessment Analytics	- Designed accuracy, precision, recall, F1-score, Cohen's Kappa, and confusion matrix analysis. - Verified cognitive balance calculations and assessment quality metrics. - Contributed to robustness evaluation and statistical interpretation of results.
Dr. A. KeerthiAssistant Professor – ECE	Quality Assurance & Validation Lead	- Conducted experimental validation using real examination question papers. - Performed expert Bloom-level verification and benchmarking. - Evaluated system robustness under ambiguous and mixed action verbs. - Assisted in system testing, performance evaluation, documentation, and demonstration preparation.